

Quantum Mechanics III

Homework 5

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Due 2026-10-04 23:59

Topic

Scattering and Measured Cross Sections

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Prove the first displayed relation used in Scattering and Measured Cross Sections. State all assumptions and conventions.

$$S = \Omega_-^\dagger \Omega_+$$

2. Calculate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$d\sigma/d\Omega = |f(\theta, \varphi)|^2$$

3. Interpret the third relation in two limiting regimes and explain the physical meaning of the result.

$$\sigma_{tot} = (4\pi/k) \text{Im} f(0)$$

Show enough intermediate work that another student can reproduce the calculation.